25900513 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEMAT202 - NUMERICAL METHODS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. Find the relative error of the number 9.7 if both of its digits are correct. [U] / [CO1]

2. Compare the false position method and the bisection method. [K] / [CO2]

3. Write the expression for the second forward difference $\Delta^2 y_i$. [U] / [CO4]

4. Define the differential operator D. [K] / [CO4]

5. Give the equation of a Lagrange polynomial of degree two passing through (x_0, y_0), (x_1, y_1) and (x_2, y_2) [U] / [CO3]

6. Write first approximation formula in the method of successive approximations. [K] / [CO4]

7. Using the following data, find $\frac{dy}{dx}$ at x = 7 [A] / [CO4]

x	1	3	5	7
y	6	11	19	27

8. Using Stirling's formula, derive the expression for the first derivative and second derivative of a function at a tabular point. [A] / [CO4]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. Find a root of the equation $x^3 - 4x - 9 = 0$ in the interval $[2, 3]$ by the bisection method. Perform 5 iterations. [A] / [CO2]

10. Solve $f(x) = x^2 - 7 = 0$ using Newton–Raphson method (two iterations). [A] / [CO2]

11. Choosing Newton's backward difference interpolation, fit a polynomial for the data $(1, 15), (3, 27)$ and $(5, 41)$. [A] / [CO3]

12. Evaluate : [A] / [CO4]

    1. $\Delta^2(x^3 + 3x + 5)$
    2. $\Delta(x^3 + 5x^2 - 2x - 10)$

13. Determine the polynomial $f(x)$ that fits the following data using Newton's divided differences: [U] / [CO3]

x	0	2	3	5
$f(x)$	1	5	10	26

14. From the following table, estimate the value of x corresponding to $y = 6$. [U] / [CO4]

y	2	4	8
x	1	2	3

15. The distance (x cm) traversed by a particle at different times (t seconds) are given below: [An] / [CO1]

t	0.0	0.1	0.2	0.3	0.4	0.5	0.6
x	3.01	3.16	3.29	3.36	3.40	3.38	3.32

Find the acceleration of the particle and estimate the errors in the value of the acceleration at $t = 0.3$ seconds.

16. Using the following data, find the area bounded by the curve, f(x) and the x-axis from x = 7.47 to x = 7.52. [A] / [CO4]

x	7.47	7.48	7.49	7.50	7.51	7.52
f(x)	1.93	1.95	1.98	2.01	2.03	2.06

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. Solve $f(x) = x^3 - 2x - 5 = 0$ using iteration method up to five iterations. [E] / [CO2]

18. Find the number of men getting wages between 10 rupees and 15 rupees by using Newton's Forward Difference Interpolation formula from the following table : [A] / [CO3]

Wages	0 - 10	10 - 20	20 - 30	30 - 40
No. of Men	9	30	35	42

19. Determine the values of y when $x = 3.2$ from the data $y(2.0) = 246.2, y(2.5) = 409.3, y(3.0) = 537.2, y(3.5) = 636.3$ and $y(4.0) = 715.9$ using (i) Gauss's forward formula (ii) Gauss's backward formula. [A] / [CO4]

20. Evaluate f(0.23) and f(0.29) from the following table by Lagrange's Interpolation Formula [E] / [CO3]

x	0.20	0.22	0.24	0.26	0.28	0.30
f(x)	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

21. Evaluate $\int_0^{0.6} e^{-x^2}\, dx$ using Simpson's 1/3 rule with h=0.1. [A] / [CO4]

22. Approximate $\int_0^2 e^x dx$ using Simpson's 3/8 rule with 6 subintervals. [A] / [CO4]

[10x3 = 30]